

Assignment 2

UCD for development of a Service Business Site



GOALS

This second assignment is meant to give you hands on experience for the User Centered Design (UCD) approach toward the development of a high-fidelity prototype of a service business site.

By “service business” I mean a type of business that offers services booked by appointment. For example, a hairdresser site, a dentist site, a bike repair site, a veterinarian site, etc.

We will work within the User-Centered Design approach which involves prototyping (iterative design) as well as taking into consideration users’ goals.

Here are examples of user’s goals:

- Bike Repair site
 - Nahar wants to know when the bike repair shop is open, where it is located and whether he can go without an appointment.
- Hairdresser site
 - Lucy wants to make an appointment for a short haircut with hairdresser Jessica on Wednesday May 27th at 2pm.
- Veterinarian site
 - Eric wants to make an appointment to bring his dog Toby to the veterinarian *GreatPets* to get Toby’s claws cut.
- Dentist site
 - Jahir wants to better understand what a root canal is and how much it costs.

By carrying out this assignment, you will:

- Better understand prototyping approaches, in particular the development of mockups with the help of tools, such as Figma.
- Practice the iterative design approach
- Continue exploring visual design as started in the last assignment
- Become familiar with HTML, CSS, Bootstrap and JavaScript for building the business web site

SEG3125 – Analysis and Design of User Interfaces



SCHEDULE / DEADLINES

Here are the important calendar events for this assignment.

- **Monday, May 25th tutorial**

Make sure to attend May 25th tutorial in which you will further explore the requirements for this assignment and the Figma software for prototyping.

- **Forum monitoring on Brightspace between Tuesday May 26th and Friday June 5th**

If something is not clear, if you need help about the assignment, then please ask questions on Brightspace forum. Make sure you get help. Any question is fine, as long as it is asked in a polite and respectful manner. TAs will be monitoring the forum regularly and answer your questions. Not only will their answers help you, they will help other students in the class.

- **Emails to TAs between Tuesday May 26th and Friday June 5th (if needed)**

ONLY if the forum help is not sufficient, you can request a one-on-one help session from the TA assigned to your lab session (make sure you contact the TA for YOUR session – Z01 to Z04).

- **Report submission: Sunday, June 7th, midnight**

You MUST submit your report, which will contain a link to your portfolio by Sunday, June 7th, midnight. *No extension will be given. The only exceptions will be for justified medical reasons with medical certificates. The penalty is 20% per day. There is a grace period till 1am... but then it's -20%!*

- **Group review registration: Sunday, June 7th, midnight**

Make sure you put your name on the group review schedule. You MUST participate in the group review to practice your analysis skills and get feedback on your design.

- **Group review participation between June 11th and June 17th**

Your scheduled group review session will be within 10 days of your submission. Not registering or missing the group review is 20% off your assignment.



ONLINE TUTORIALS / TECHNOLOGIES

In addition to what your TA will present in his tutorial, you can consult the following resources.

Mockups

For the development of your mockups, please consult the following tutorials:

- Figma Tutorial
 - <https://www.youtube.com/watch?v=eZldKx-jPag>
 - <https://www.youtube.com/watch?v=JGLfyTDgfDc>
- Pencil Project Tutorial
 - <https://www.youtube.com/watch?v=wT64vMT1Ax0>

Web development technologies

For the development of your final prototype, in addition to the tutorials about web development posted in the Assignment 1 requirement document, please see below for tutorials related to React*:

- <https://react-bootstrap.netlify.app/docs/getting-started/introduction/>
- <https://www.youtube.com/watch?v=acgtyCnQAAw>
- <https://www.youtube.com/watch?v=SqcY0GIETPk>
- <https://www.youtube.com/watch?v=x4rFhThSX04>

* Using JavaScript is mandatory, but using React (at this time in the semester) is not yet mandatory. However, it is with React that TAs can help you. You can use a different JavaScript framework, but without support from the SEG3125 team.



REQUIREMENTS

1. Decide on a service site

Decide on which service business you want to explore. In the GOAL section above, I suggested: hair salon, dentist, bike repair, veterinarian. But those are only suggestions. Any service business is fine.

You will be asked (in section 2 below) to define personas with goals. In addition to providing the necessary functionalities to meet the personas' goals, your service site must, at a minimum, allow a user to:

- Find out the name and type of your business
- Find out contact information about your business
- See information about the site designer (e.g. a “designed by...” information somewhere on the page)
- Stay interested in your services and want to use them (challenge for your visual design)

2. Create 2 personas

Develop *two personas* as target users of the Service Web Site. For EACH persona:

- Provide a name to your persona
- Provide 3 intrinsic characteristics to your persona
- Provide the relation to technology and relation to the domain of your persona
- Provide 1 goal to your persona (e.g. booking an appointment, finding out information about different services and prices, etc)

The chosen goals might vary slightly depending on your type of service, but since I'm asking for a service site, goals should be around getting access to those services.

3. Create 2 visually different storyboards

A storyboard is a series of mockups showing how a persona will linearly (step by step) achieve his/her goal. For each of your persona, create a storyboard. Each storyboard should contain at least three mockups showing 3 steps for achieving a goal.

For example, if the goal is booking an appointment for a short haircut at the hairdresser, the storyboard should contain:

- Mockup with service selection (short hair) and access to the reservation
- Mockup showing a calendar for booking
- Mockup showing appointment confirmation

To practice generating alternatives and continue our exploration of visual design principles, you have to create visually different mockups for your two storyboards. Explore two different color themes, different layouts, different ways of displaying the information. Make sure your storyboards are visually distinct from each other and that they reflect possible personalities of your site.

PLEASE NOTE: You will show your storyboards in the group reviews. Please make sure the designs are different, as your colleagues will be asked to compare them and give feedback.

4. Create one high-fidelity interactive prototype

Using a *JavaScript framework (e.g., React)*, create a single high-fidelity interactive prototype that will combine the goals of your two personas and reflect your visual design choices.

The final design choice can be a combination of the choices made in the two storyboards or could be based on a single storyboard. Make sure you explain the relationship between your storyboards and your final prototype in your report.

The high-fidelity prototype should meet the following technical requirements:

- Be developed using a JavaScript framework (e.g., React)
- Be interactive (respond to commands – buttons, list selections)
- Be accessible through your portfolio (from assignment 1)

Your high-fidelity prototype should be guided by visual design principles:

- The use of a color theme to insure consistency
- A reflection on typography
- A reflection on iconography
- A reflection on layout and the use of negative space
- A reflection on the principles of contrast, scale, balance, visual hierarchy

5. Link your portfolio to your high-fidelity prototype

You created your portfolio during Assignment 1. You must now link your portfolio to the high-fidelity prototype of the service website. You can also (if you wish) link your portfolio to your two storyboards to demonstrate your working method and contribute to the expertise (*how you work*) section of your portfolio.

Your portfolio and service site should be accessible to the public (or, if you prefer, password protected with the password provided in your report).

The TA must be able to test your UI without having to communicate with you. This is ESSENTIAL.

6. Write a short report describing your design process and choices:

Your report should include:

1. Designer: Name, Student Number
2. Service: the name and type of service you decided to do
3. Storyboard with mockups:
 - a. Definition of your 2 personas
 - b. Your 2 storyboards (with at least 3 mockups each), one for each persona, and both visually different
4. High-fidelity prototype
 - a. An explanation of the visual design choices you made in relation to the storyboards (differences, similarities)
 - b. Link to your portfolio which will contain a link to the service site
5. Code: A link to Github (or other repository containing your code)
6. Generative AI Acknowledgement: Many of you will use generative AI to help you create your (a) mockups, (b) high-fidelity prototypes and (c) your report. Describe the role of generative AI for each of those, providing the GenAI tools used and your interactions with them.



EVALUATION

This assignment is worth 8% of your semester and is evaluated on 100 points.

- Your report's content and clarity (10 points)
 - Report easy to read and answers the required questions.
 - Go through the list of 6 elements given above. Your report should contain all these elements with answers to better understand your design choices and the help provided by GenAI.
- Your two personas + storyboards (20 points)
- Your hosted web site, accessible, made according to the requirements provided (50 points)
 - UI should reflect the choices made visual design
 - UI should be an extension/combination of storyboards
- Your active participation in the group review (20 points)
 - Ability to analyze other student's design according to the principles explored in the lecture
 - Ability to answer questions from TA and other students



SUBMISSION CONTENT / PLATFORM

Each student (it is an individual assignment) must submit their report, PDF version (see all its requirements above) on Brightspace. The link can be found in the Assignment section, for Assignment 2.

ALSO, very important, **MAKE SURE you register on the signup sheet for your group review.** You will be informed when the signup sheet is available.



QUESTIONS

- See the section Schedule/Deadline of the current document. It indicates tutorials and forum for help and questions.
- Please DO NOT send the professor any emails. The forum help should be enough to obtain answers to your questions. Do not be shy to ask questions on the forum. Any answers provided there by the TA will help many other students.
- If the forum help is not sufficient, email the TA assigned to your lab session.